

Syed Taha

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Education

Institute of Business Administration (IBA), Karachi

Expected May 2027

Bachelor of Science in Computer Science

GPA: 3.53/4.0

Relevant Coursework: Natural Language Processing, Machine Learning, Computer Architecture, Operating Systems, Parallel and Distributed Computing, French 1.

Technical Skills

Programming Languages:	C++, Python, RISC-V Assembly, Bash, JavaScript
High-Performance Computing:	CUDA (Basic), Vectorization, Multi-threading, Performance Profiling
Scientific Computing & ML:	TensorFlow, PyTorch, NumPy, Pandas, Matplotlib, Scikit-learn
Systems & Development Tools:	Linux, Git, GitHub Actions, LaTeX

Research & Technical Projects

WebGL2 Path Tracer

Personal Project

JavaScript, GLSL, WebGL2, Web Audio API, Three.js Loaders

- Built a physically-based path tracer from scratch in WebGL2 (GLSL ES 3.0), implementing Monte Carlo sampling, GGX microfacet BRDF, Fresnel reflectance, and multi-bounce light transport with Russian roulette termination.
- Engineered a real-time underwater rendering system featuring Snell's Window, total internal reflection, Beer's law wavelength-dependent absorption, analytical caustics via wave-surface Laplacian, and animated water normals computed with central differences.
- Implemented a full day/night sky model with Voronoi-based star field, procedural moon, and smooth dusk/dawn transitions; extended with equirectangular HDRI/EXR environment map support via dynamically imported Three.js loaders.
- Designed a denoising pipeline using an A-Trous wavelet filter with normal-guided edge stopping, enabling usable image quality at low sample counts.
- Architected the codebase as an OOP ES module system (**Camera, Scene, Renderer, Sun, InputHandler, UI**) with clear separation of concerns between GL state, scene data, and DOM interaction.
- Added a procedural deep-ocean soundscape using the Web Audio API — pink noise (Voss-McCartney), LFO-modulated low-pass swell, bandpass wind, and stochastic pitch-descending gulp synthesis.

LLM Inference Optimization in the xv6 Educational OS

Research Project

C, RISC-V, xv6, Systems Programming, LLaMA-2

- Engineered a POSIX-compliant Shared Memory (SHM) subsystem for the xv6 kernel to enable zero-copy, persistent caching of Large Language Model weights (~**60MB**), eliminating redundant network transfers and reducing subsequent inference latency from ~**60s** to sub-second.
- Designed a cycle-accurate performance profiling library that tracks call hierarchies and identified matrix multiplication as the dominant bottleneck (consuming **87-92%** of inference time), providing a data-driven basis for optimization.
- Successfully ported the LLaMA-2 inference engine to xv6, resolving system-level challenges (stack overflow via iterative quicksort) and replacing Linux dependencies, achieving stable throughput of ~**16 tokens/sec** on RISC-V.

Vectorized CNN Inference on 32-bit RISC-V Cores

Research Project

RISC-V Assembly, RVV Vector Extensions, CNN, Edge Computing

- Implemented a complete Convolutional Neural Network in **RISC-V assembly**, leveraging the Vector Extension (RVV) to optimize core operations (convolution, pooling, softmax) for edge computing constraints.
- Achieved a **7.3x speedup in total inference time** (8.223s to 1.122s) and a **12.2x reduction in convolution instructions** via algorithmic redesign (im2col + GEMM) and FP16 compression.
- Authored a full research paper detailing the methodology and benchmarks.

High-Performance Othello AI with MCTS & Optimized CNNs

Research Project

Python, PyTorch, Monte Carlo Tree Search, Deep Reinforcement Learning

- Researched and developed a deep reinforcement learning AI by integrating Monte Carlo Tree Search (MCTS) with optimized Convolutional Neural Networks for board evaluation and move prediction.
- Systematically optimized the CNN architecture, reducing model size by **50%** (85MB to 42.66MB) while

increasing gameplay speed **10x** (1000ms to 100ms per move) and maintaining a **>90%** win rate against larger baselines.

- Conducted rigorous internal tournaments with a custom scoring metric balancing wins, speed, model size, and search depth to guide architectural and hyperparameter decisions.

Kaggle Competition: Enhanced Safe Driver Prediction

1st Place

Python, XGBoost, LightGBM, CatBoost, Scikit-learn

- Achieved **1st place** (Private Leaderboard AUROC: 0.64671) by developing a stacking ensemble of gradient boosting models (XGBoost, LightGBM, CatBoost) with a logistic regression meta-learner.
- Conducted systematic model evaluation and comprehensive data exploration, informing a refined preprocessing strategy that handled extensive missing data and removed non-predictive features.
- Demonstrated expertise in ensemble methods, feature engineering, and highly imbalanced tabular data.

Multi-Colony Population Dynamics – Agent-Based Simulation

Research Project

Python, Q-Learning, Particle Swarm Optimization, Agent-Based Modeling

- Built a grid-based multi-agent simulation environment where competing colonies of agents forage for resources, engage in combat, expand territory, and reproduce — comparing three algorithmic strategies: Greedy heuristics, decentralized Q-Learning, and a modified PSO variant.
- Designed a novel colony-fitness-guided PSO variant that replaces personal and global best positions with a local hill-climbing heuristic, allowing agents to continuously optimize based on collective colony fitness without explicit positional memory.
- Implemented a multi-objective fitness function incorporating territorial control, resource accumulation, and agent cohesion, and analyzed emergent behaviors including grouping, stagnation, and inter-colony aggression across hundreds of simulation trials.
- Found that Greedy agents dominated early through aggressive territorial expansion but suffered long-term resource depletion; Q-Learning agents showed stronger individual adaptability; and PSO performance was highly sensitive to parameterization, with inertia and velocity clamping being critical factors.

Modular 2D Physics & Orbital Mechanics Engine

Personal Project

- Architected and built a modular 2D physics engine from scratch in C++, implementing rigid body dynamics, velocity-based movement, and AABB collision detection/resolution.
- Extended the core engine into an orbital mechanics simulation to model gravitational N-body interactions, applying vector mathematics and numerical integration.
- Developed a real-time visualization layer using SFML, creating an interactive tool to validate physical simulations.

Chess Engine with Minimax AI

Personal Project

C++, Minimax Algorithm, Multithreading, Bash Scripting, Git

- Architected and built a fully functional chess engine from scratch in C++, implementing core game logic, move generation, and a Minimax-based AI opponent — independently, with only prior OOP knowledge and self-directed learning.
- Engineered a multithreaded architecture to offload AI computation to a background thread, preventing UI blocking and ensuring a responsive user experience during deep search evaluations.
- Developed proficiency in systems-level tooling throughout the project, including Bash scripting, version control with Git/GitHub, and low-level debugging — acquiring skills spanning AI, concurrency, and software engineering workflows without formal coursework.

Professional Experience

Teaching Assistant - Computer Architecture

Jan 2026 – May 2026

Institute of Business Administration, Karachi

- Mentored 10 student teams to successfully complete RISC-V assembly and hardware-optimized projects on S4 models for galaxy morphology classification and edge computing.
- Leading one student team post-course completion toward the publication of their research findings.
- Guided students with a strong focus on formal methods, hardware verification, and technical report writing.
- Graded assignments and assessments while maintaining consistent academic standards across course deliverables.

Teaching Assistant - Introduction to Programming & OOP

Aug 2024 – Dec 2025

Institute of Business Administration, Karachi

- Facilitated weekly lab sessions for 100+ students, providing technical guidance on problem-solving, debugging, and core programming concepts in C++.

- Acted as primary liaison between faculty and students, clarifying course requirements and ensuring consistent academic standards.
- Developed and organized lab materials, supporting continuous improvement in course delivery and student engagement.

Executive Council - Program Design

Aug 2024 – May 2025

IBA Data Science Society

- Designed and executed technical workshops and large-scale events, including a Datathon x Datafest with 400+ registrations in collaboration with Google Developer Groups on Campus (GDGoC)
- Structured program modules and event content to maximize participant learning outcomes and technical skill development.

Software Engineering Fellow

Jul 2024 – Sept 2024

Headstarter AI

Remote

- Delivered 5 AI-powered web apps in 5 weeks using React/Next.js, Firebase, Clerk, and Vercel under weekly agile sprints with CI/CD—scaled each to 200+ users and launched an IBA teacher-review portal with 300 waitlist sign-ups.
- Co-developed a Next.js customer-support agent with a custom OpenAI+Pinecone RAG pipeline and a GPT-4o-powered flashcard SaaS; participated in weekly innovation sessions with engineers from Google, Y Combinator, Stanford, Amazon, and top startups.

Achievements & Competitions

- **Kaggle 1st Place:** Enhanced Safe Driver Prediction competition (outperformed 100+ participants).
- **Competitive Programming:** 1st place in “Hello World” Competitive Programming Competition (Advanced Category), organized by IBA Computer Science Society.
- **Comprehensive System Building:** Successfully ported and optimized a modern LLM inference pipeline to a minimal educational operating system (xv6).